Wisdom is a personality trait that is composed of self-reflection, pro-social behaviors, emotional regulation, acceptance of diverse perspectives, decisiveness, social advising, and spirituality. Wisdom is separable from intelligence. Greater levels of wisdom are linked to better overall health, well-being, happiness, life satisfaction, and resilience. However, the underlying genetic architecture of wisdom and its genetic relationships with health and cognition remain unexplored. We collected responses to SD-WISE-7, a validated survey measure of wisdom, in 131,870 23andMe research-consented participants and conducted the first large-scale genome-wide association study (GWAS) of this trait. We identified one genome-wide significant locus (rs1028640, P=3.7e-08) near the MANEA gene on chromosome 6. Wisdom was heritable (h2SNP=0.03±0.004) and genetically correlated with numerous complex traits, including subjective well-being (rg=0.34, P=1.52e-09), neuroticism (rg=-0.38, P=1.87e-16), loneliness (rg=-0.27, P=8.45e-07), other personality traits such as openness (rg=0.41, P=1.37e-07) and risk tolerance (rg=0.36, P=6.94e-11). Crucially, wisdom was genetically distinct from intelligence (rg=0.036, P=0.39) and cognitive educational attainment (rg=-0.01, P=0.83). We calculated wisdom polygenic scores (PGS) in All of Us (N≈200,000) and identified significant associations with psychosocial outcomes (e.g., social satisfaction ratings P=1.16e-12 and subjective mental health P=5.93e-10), but not cognitive performance (p=xx). This study advances our understanding of wisdom as a biologically-grounded trait that could ultimately be used to develop informed interventions that increase wisdom and improve health and well-being.